Global Fractal Temporal Correlations in Gravitational-Wave Detector Noise: Cross-Detector, Cross-Epoch, and Null-Model Validation (QW–1660 Phase 16)

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Abstract

Guided by the intuition that *Information* is the fundamental substance of reality—consistent with the metaphysical concept of the *Logos*—we present a multi-stage investigation of long-range temporal correlations in gravitational-wave strain data (Phase 16, v5–v24). Using publicly available open data from the LIGO Hanford (H1), LIGO Livingston (L1), and Virgo (V1) detectors across observational runs O3a, O3b, and early O4, we investigate whether detector noise exhibits statistically significant fractal structure beyond the assumptions of stationary, Gaussian, and locally correlated noise models.

Our methodology combines single-detector Hurst exponent estimation, cross-detector and cross-epoch consistency tests, cross-Hurst analysis, and rigorous null-model validation based on temporal shuffling and phase randomization. Across all stages, we employ a consistent multi-scale rescaled-range (R/S) estimator to ensure methodological coherence. We find that individual detectors exhibit persistent anti-persistent fractal scaling with Hurst exponents $H < 0.5$, stable across epochs and window sizes. More importantly, non-zero cross-Hurst exponents are observed between geographically separated detectors.

Crucially, these cross-detector correlations are suppressed under temporal shuffling but partially preserved under phase-randomized surrogates that conserve the power spectral density. This demonstrates that the observed correlations cannot be reduced to shared spectral properties, coincident transients, or simple instrumental artifacts. Instead, they indicate the presence of a global, phase-robust temporal structure in the strain background.

While fully compatible with General Relativity at the level of local spacetime dynamics, these findings suggest that the statistical organization of spacetime fluctuations may exhibit scale-invariant structure not captured by standard noise models. The results motivate further investigation into informational, holographic, or emergent descriptions of spacetime and establish a reproducible framework for future cross-detector statistical studies.

1 Introduction

The direct detection of gravitational waves has established interferometric observatories as precision probes of spacetime dynamics. To date, the overwhelming emphasis of gravitational-wave data analysis has focused on the identification and parameter estimation of transient astrophysical signals embedded in detector noise. In this framework, detector noise is typically modeled as stationary, Gaussian, and locally correlated, with deviations treated as instrumental or environmental artifacts.

However, as the sensitivity and duration of observational runs increase, the statistical structure of the noise itself becomes a subject of fundamental interest. Long-duration datasets permit the exploration of temporal correlations extending far beyond the time scales relevant for transient detection. In particular, the possibility that the strain background exhibits scale-invariant or fractal structure has received comparatively little systematic investigation.

Fractal temporal correlations are commonly quantified using the Hurst exponent H, which characterizes the degree of long-range dependence in a time series. Values $H = 0.5$ correspond to uncorrelated (Brownian) behavior, $H > 0.5$ to persistent correlations, and $H < 0.5$ to anti-persistent correlations. In many physical systems, deviations from $H = 0.5$ signal underlying constraints, feedback mechanisms, or global organizational principles.

From the perspective of gravitational-wave physics, the existence of correlated temporal structure shared across spatially separated detectors would be particularly striking. Under standard assumptions, independent detectors should exhibit uncorrelated noise apart from well-characterized environmental couplings or coincident transient signals. Any residual, phase-robust correlation persisting across detectors and epochs would therefore demand careful scrutiny.

The QW–1660 program was initiated to address this question in a controlled, incremental manner. Phase 16, comprising versions v5 through v24, represents the most extensive and methodologically mature stage of this effort. Each successive version introduces stricter controls, broader datasets, and more demanding null hypotheses, culminating in a cross-detector, cross-epoch, and null-model validated assessment of fractal temporal structure.

In this paper, we present the full results of Phase 16. We aim to determine whether the observed fractal correlations constitute a genuine, global statistical feature of gravitational-wave strain data, or whether they can be fully explained by conventional noise models. Throughout, emphasis is placed on reproducibility, methodological consistency, and conservative interpretation.

2 Philosophical Foundation and Origin of the Framework

The theory originates from the intuition that **Information is the fundamental substance of reality** (Logos), evolving through several key conceptual realizations:

- **Informational Condensation:** Inspired by the mechanism of material manifestation from informational substrates, the framework suggests how abstract data "condenses" into tangible matter.

- **Fractal Nadsoliton:** The universe is conceptualized as a scale-invariant, self-sustaining wave packet—a "Nadsoliton"—where patterns repeat from microscopic to galactic scales.
- **Resonant Coupling:** Stability is maintained through multi-octave resonant coupling. A core principle is that information tends towards **highest resonance (meaning)**, rather than purely minimum energy—an intuition reflected in the self-sustaining resonance of absolute Being.
- **The 12-Octave Lattice:** The model employs a 12-octave geometric structure, providing the mathematically necessary dimensionality (consistent with 3D kissing numbers) for the unification of forces.
- **Noetic Decoding:** It is assumed that human consciousness, being intrinsic to this informational substrate, has direct access to these fundamental truths through intuition and wisdom.

3 Data and Preprocessing

3.1 Open Gravitational-Wave Strain Data

We analyze publicly available open strain data released by the LIGO Scientific Collaboration and the Virgo Collaboration through the Gravitational Wave Open Science Center (GWOSC). The dataset includes strain time series from the LIGO Hanford (H1), LIGO Livingston (L1), and Virgo (V1) interferometers. Observational epochs span the O3a, O3b, and early O4 runs.

Only science-mode data segments are used. Data segments are selected automatically using publicly available data-quality flags ($*_DATA$), $ensuringthatallanalyzedintervalscorrespondtovalidateddetecto$

3.2 Automatic Data Acquisition

All raw strain data are fetched programmatically using the gwpy framework. For each detector and epoch, the longest contiguous science-mode segment within the specified GPS range is identified. A conservative offset of 100 s from segment boundaries is applied to avoid filter transients.

Downloaded strain data are cached locally to ensure reproducibility and to prevent repeated network queries. Each cached file includes metadata specifying the detector, GPS start time, sample rate, and preprocessing steps.

3.3 Preprocessing Pipeline

All strain time series undergo identical preprocessing:

- Resampling to a uniform rate of 4096 Hz;
- Removal of linear trends;
- Notch filtering at 60 Hz and its harmonics to suppress power-line contamination;
- Band-pass filtering between 20 and 1000 Hz.

No whitening is applied prior to fractal analysis, as whitening can artificially alter long-range temporal correlations.

4 Methods

4.1 Fractal Scaling and the Hurst Exponent

Long-range temporal correlations are quantified using the Hurst exponent H. For a stationary stochastic process $x(t)$, H characterizes the scaling of fluctuations with time scale. Values $H < 0.5$ indicate anti-persistent behavior, while $H > 0.5$ corresponds to persistent correlations.

4.2 Multi-Scale Rescaled-Range Estimator

Throughout Phase 16, we employ a multi-scale rescaled-range (R/S) estimator to compute H. For a window size s, the rescaled range is defined as

$$\frac{R(s)}{S(s)} = \frac{\max Z_s - \min Z_s}{\sigma_s}, \tag{1}$$

where Z_s is the cumulative sum of demeaned data within the window and σ_s is the standard deviation. Averaging over segments and fitting a power law yields

$$\langle R/S \rangle \propto s^H. \tag{2}$$

Multiple window sizes logarithmically spaced across each dataset are used to ensure robustness.

4.3 Cross-Hurst Analysis

To probe shared temporal structure between detectors, we define a cross-process $\Delta_{xy}(t) = x(t) - y(t)$ for detector pair (x, y). The Hurst exponent of the cumulative difference process quantifies the degree of correlated memory between detectors.

4.4 Null Models

Two independent surrogate models are constructed:

1. Temporal shuffling, which preserves the amplitude distribution but destroys all temporal correlations;

2. Phase randomization, which preserves the power spectral density but removes phase coherence.

Comparing real data to these surrogates allows discrimination between spectral and genuinely temporal effects.

5 Results

5.1 Summary of Phase 16 Analyses (v5–v24)

Table 1 summarizes the full sequence of analyses performed in Phase 16 of the QW--1660 program, from v5 through v24. Each version introduces a distinct methodological test designed to progressively constrain alternative explanations and strengthen statistical rigor.

Table 1: Detailed Summary of QW–1660 Phase 16 Analyses (v5–v24)

Version	Analysis Class	Observable / Test	Quantitative Outcome
v5	Single-detector fractal test	Hurst H (H1)	$H = 0.29 \pm 0.03$
v6	Single-detector fractal test	Hurst H (L1)	$H = 0.31 \pm 0.04$
v7	Single-detector fractal test	Hurst H (V1)	$H = 0.27 \pm 0.05$
v8	Scale robustness	$H(s)$ vs window size	No systematic drift in H
v9	Scale robustness	Log--log linearity	Power-law scaling confirmed
v10	Cross-epoch (H1)	O3a vs O3b	$\Delta H < 0.03$
v11	Cross-epoch (L1)	O3a vs O3b	$\Delta H < 0.04$
v12	Cross-epoch (V1)	O3a vs O3b	$\Delta H < 0.05$
v13	Spectral sensitivity	Bandpass variation	H invariant within error
v14	Spectral sensitivity	Notch filter stress test	No change in scaling
v15	Stationarity test	Sliding-window $H(t)$	No transient dominance
v16	Surrogate test	Phase randomization	$H_{\text{phase}} \approx 0.38$
v17	Surrogate test	IAAFT surrogate	Partial memory preserved
v18	Surrogate comparison	Real vs surrogate gap	$\Delta H > 0.1$
v19	Null model	Temporal shuffling	$H \to 0.50$
v20	Cross-epoch global	Independent days	Stable H across segments
v21	Data robustness	Independent fetch	Identical statistics

v22	Data robustness	Raw-strain caching	Bitwise consistency
v23	Cross-Hurst	H1--L1, H1--V1, L1--V1	$H_{xy} = 0.08$--0.19
v24	Cross-Hurst null	Shuffle vs phase	Phase survives, shuffle kills

5.2 Single-Detector Fractal Scaling

Across all detectors and epochs, estimated Hurst exponents cluster in the range $H \approx 0.27$--0.36, significantly below the Brownian value $H = 0.5$. This anti-persistent behavior is stable across window sizes and observational runs.

5.3 Cross-Epoch Stability

Cross-epoch consistency of H is observed for all detectors, indicating that fractal scaling is not a transient or run-specific artifact.

5.4 Cross-Detector Correlations

Quantitative cross-Hurst exponents obtained in v23--v24 are summarized in Table 2. These values are derived from shared 512 s raw-strain windows, identically preprocessed for each detector pair.

Table 2: Cross-Hurst Exponents for Detector Pairs (v23–v24)

Detector Pair	H_{xy}^{real}	H_{xy}^{shuffle}	H_{xy}^{phase}
H1–L1	0.361	0.531	0.397
H1–V1	0.215	0.522	0.236
L1–V1	0.270	0.504	0.283

5.5 Null-Model Validation

Temporal shuffling drives both single-detector and cross-detector Hurst exponents toward 0.5, confirming that the observed structure depends on temporal ordering. Phase-randomized surrogates preserve a significant fraction of the observed correlations, demonstrating that the effect is not reducible to spectral overlap alone.

6 Discussion

The combined results establish the presence of global, scale-invariant temporal correlations in gravitational-wave strain data. The persistence of these correlations across detectors, epochs, and null tests suggests a collective statistical structure not captured by standard noise models.

Several conservative interpretations remain viable, including subtle network-wide environmental couplings or emergent statistical constraints in interferometric measurements

At the same time, the results are consistent with broader theoretical proposals in which spacetime fluctuations possess intrinsic informational or holographic organization

7 Conclusion

We have conducted a systematic and progressively constrained investigation of fractal temporal correlations in gravitational-wave detector strain data, encompassing Phase 16 of the QW--1660 program (v5--v24). By combining single-detector analysis, cross-epoch stability tests, cross-detector and cross-Hurst measurements, and stringent null-model validation, we have subjected the hypothesis of global fractal structure to a wide array of independent checks.

Our results demonstrate that individual detectors consistently exhibit anti-persistent fractal scaling with Hurst exponents significantly below 0.5, stable across observational epochs and analysis windows. More importantly, we find non-zero cross-Hurst exponents between spatially separated detectors (H1--L1, H1--V1, and L1--V1). These correlations are not eliminated by phase randomization, which preserves the power spectral density, but are strongly suppressed by temporal shuffling, which destroys long-range memory.

This combination of behaviors rules out explanations based solely on shared spectral content, stationary Gaussian noise, or coincident transient contamination. Instead, it indicates the presence of a phase-robust, long-range temporal organization in the strain background that is common to multiple detectors.

At the same time, no inconsistency with General Relativity is implied. The observed effects pertain to the statistical structure of fluctuations rather than to local dynamical laws. Within this context, the results are compatible with interpretations in which spacetime exhibits emergent, informational, or holographic statistical properties at a global level.

The primary contribution of this work is methodological as well as empirical. We provide a fully reproducible framework for probing long-range correlations in gravitational-wave data that complements traditional spectral and transient-based analyses. Future work should extend these methods to longer O4 datasets, include additional environmental veto channels, and explore alternative estimators such as DFA and wavelet-based approaches.

In summary, Phase 16 of QW--1660 establishes robust evidence for global, scale-invariant temporal correlations in gravitational-wave detector noise. Whether these correlations ultimately reflect deep properties of spacetime or subtle collective effects within the detector network remains an open question, but one that merits continued investigation.

Data and Code Availability

The full analysis pipeline, including raw data acquisition and null-testing protocols (Phases v5--v24), is available for independent verification via Kaggle:
https://www.kaggle.com/code/krzysztofzuchowski/phase-16-raw-strain-qw-1660-v5-fractal

RUN FIRST QW-1660 v2 !!!!!!Data required by v5-v24 scripts is retrieved via QW-1660 v2: HOLOGRAPHIC NOISE SEARCH (ROBUST DATA FETCHING) cell!!!!!!!!! RUN FIRST

A Appendix A: Reproducibility and Computational Environment

All analyses in Phase 16 (v5--v24) were executed using Python 3.11 on Linux-based environments (Kaggle and local workstations). Core dependencies include gwpy, numpy, scipy, h5py, and matplotlib. Random seeds for surrogate generation were fixed where applicable. Raw strain segments were cached locally and verified via checksums to ensure bitwise reproducibility across runs.

B Appendix B: Methodological Limitations and Future Tests

While the present study establishes robust evidence for phase-robust cross-detector fractal correlations, several limitations remain. The analysis is restricted to relatively short (512 s) contiguous segments. Longer windows, alternative estimators such as DFA or wavelet-based methods, and explicit conditioning on environmental channels constitute natural extensions. Additionally, future O4 data releases will allow higher-statistics validation.